2 EXP1 OBJ CODE AND OUTPUT

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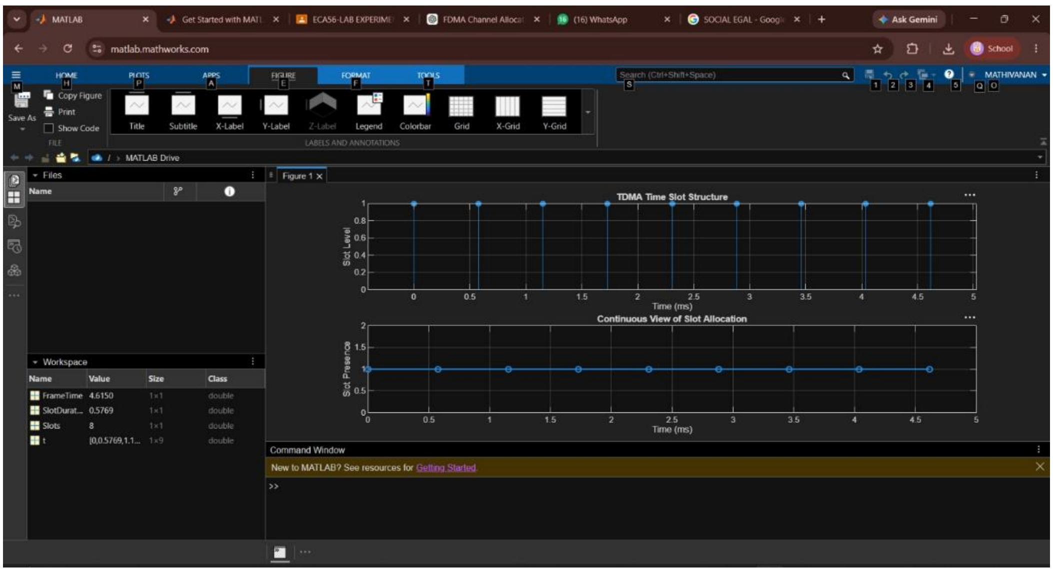
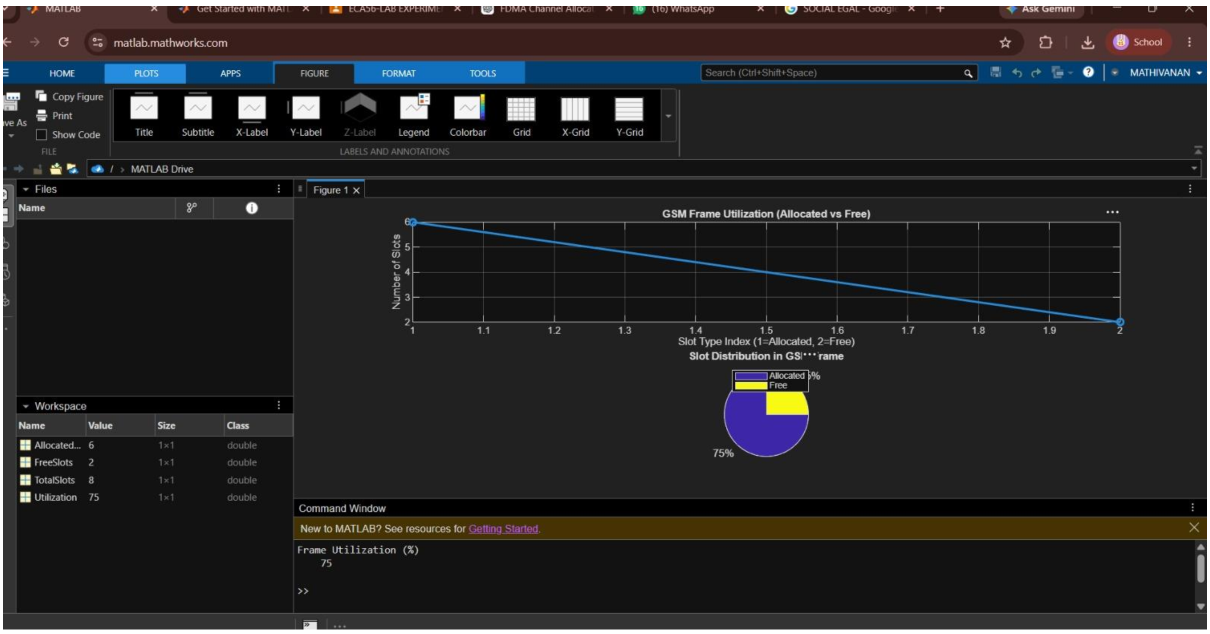
```
clc; clear; close all;

FrameTime = 4.615; Slots = 8;

SlotDuration = FrameTime/Slots;


t = 0:SlotDuration:FrameTime;

figure

subplot(2,1,1)

stem(t,ones(size(t)),'filled')

title('TDMA Time Slot Structure')

xlabel('Time (ms)')

ylabel('Slot Level')

grid on

subplot(2,1,2)

plot(t,ones(size(t)),'-o','LineWidth',1.5)

title('Continuous View of Slot Allocation')

xlabel('Time (ms)')

ylabel('Slot Presence')

grid on
```



EXP 2 OBJ 2

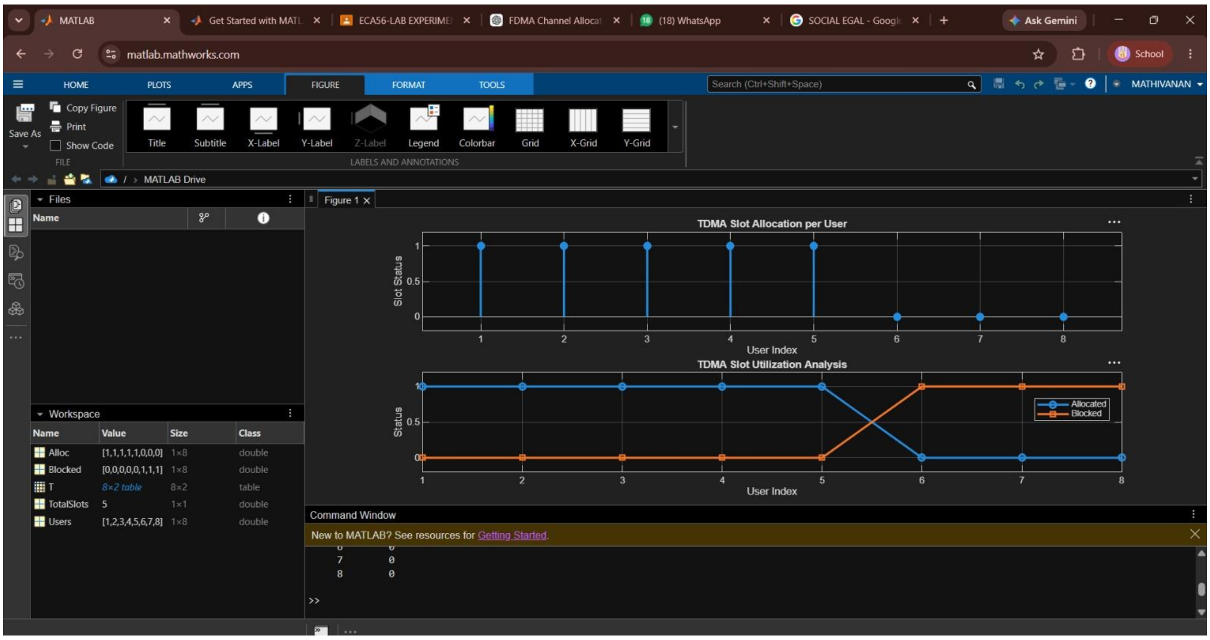
```
clc;

clear;

close all;


% Parameters

Users = 1:8;         % Total users

TotalSlots = 5;      % Available slots


% Initialize allocation

Alloc = zeros(1, length(Users));

Alloc(1:TotalSlots) = 1;     % Allocate first 5 users


% Blocked users

Blocked = 1 - Alloc;


% Display allocation table

disp('User Wise Slot Allocation');

T = table(Users', Alloc', ...

    'VariableNames', {'User', 'Status'});

disp(T);


% Create figure

figure;


%---------------- Graph 1 ----------------%

subplot(2,1,1);

stem(Users, Alloc, 'filled', 'LineWidth', 2);

title('TDMA Slot Allocation per User');

xlabel('User Index');
```

```
ylabel('Slot Status');

ylim([-0.2 1.2]);

xticks(Users);

grid on;


%---------------- Graph 2 ----------------%

subplot(2,1,2);

plot(Users, Alloc, '-o', 'LineWidth', 2);

hold on;

plot(Users, Blocked, '-s', 'LineWidth', 2);

hold off;


title('TDMA Slot Utilization Analysis');

xlabel('User Index');

ylabel('Status');

legend('Allocated', 'Blocked', 'Location', 'best');

ylim([-0.2 1.2]);

xticks(Users);

grid on;
```



EXP 2 OBJ 3

```
clc; clear; close all;

d = 0.1:0.1:5; % Distance (km)
f = [900 3500 28000]; % Frequency (MHz)
PL1 = 32.44 + 20*log10(f(1)) + 20*log10(d);
PL2 = 32.44 + 20*log10(f(2)) + 20*log10(d);
PL3 = 32.44 + 20*log10(f(3)) + 20*log10(d);
figure
% -------- Graph 1 --------
subplot(2,1,1)
plot(d,PL1,'b','LineWidth',2); hold on
plot(d,PL2,'r','LineWidth',2)
plot(d,PL3,'k','LineWidth',2)
title('Propagation Loss vs Distance')
xlabel('Distance (km)')
ylabel('Loss (dB)')
grid on
% -------- Graph 2 (NO legend used to avoid error) --------
subplot(2,1,2)
Freq = [900 3500 28000];
Loss = [PL1(10) PL2(10) PL3(10)];
plot(Freq,Loss,'-o','LineWidth',2)
title('Propagation Loss vs Frequency')
xlabel('Frequency (MHz)')
ylabel('Loss (dB)')
grid on
```


TDMA Slot Allocation per User
Slot Status
User Index
TDMA Slot Utilization Analysis
Status
User Index
Allocated
Blocked
Command Window
New to MATLAB? See resources for Getting Started.